

COURSE CODE/TITLE: CPE212
Automating Server Management

SECTION: CPE31S2

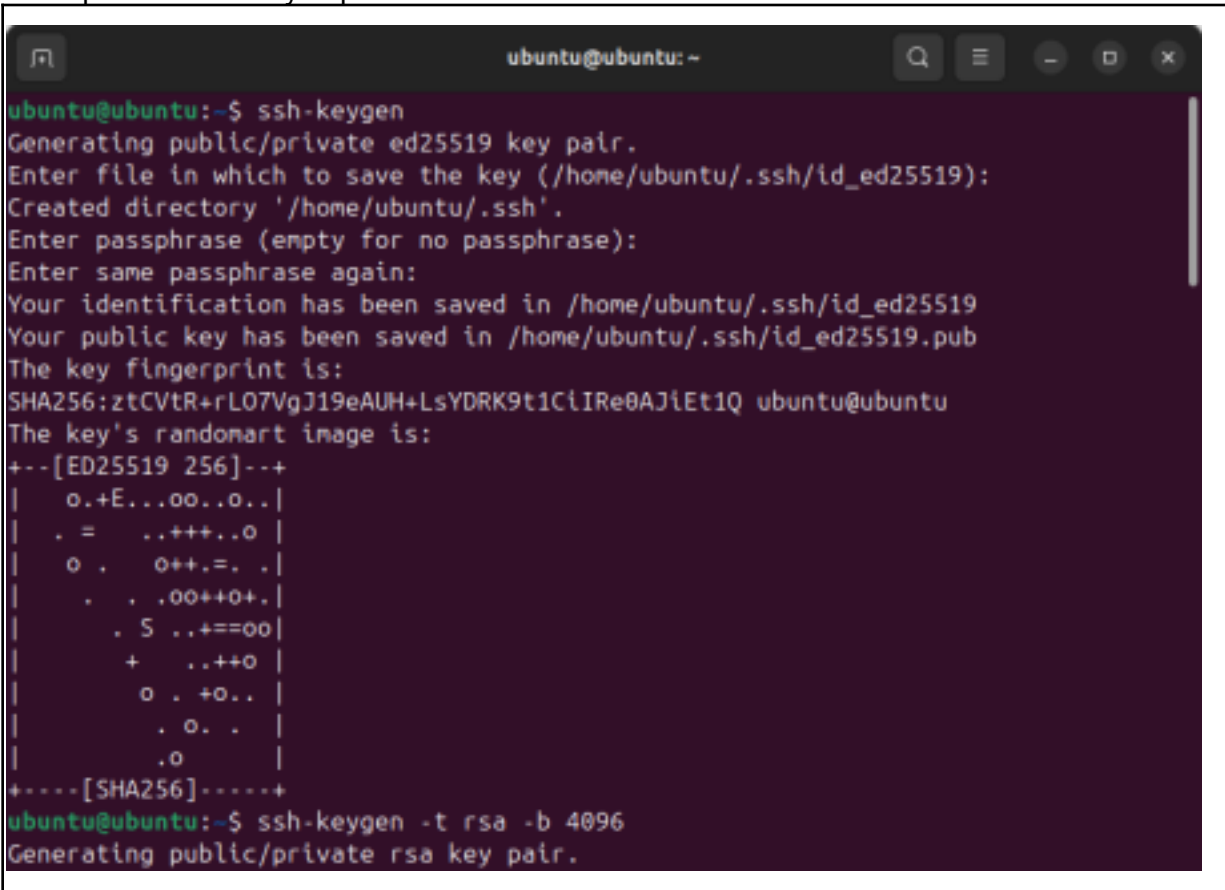
DATE PERFORMED: 07/30/2026 **DATE SUBMITTED:** 07/30/2026 **NAME:** Melquisedec

Iquin **INSTRUCTOR:** Engr. Jimlord Quejado

ACTIVITY NO. 2.1 SSH Key-Based Authentication and GIT Setup

1. PROCEDURE OUTPUT

What procedures did you perform?



```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ ssh-keygen  
Generating public/private ed25519 key pair.  
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):  
Created directory '/home/ubuntu/.ssh'.  
Enter passphrase (empty for no passphrase):  
Enter same passphrase again:  
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519  
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub  
The key fingerprint is:  
SHA256:ztCVtR+rL07VgJ19eAUH+LsYDRK9t1CtIRe0AJiEt1Q ubuntu@ubuntu  
The key's randomart image is:  
+--[ED25519 256]--+  
|  o.+E...oo..o..|  
| . = ..+++..o |  
| o .  o++.=. .|  
| . . .oo++o+.|  
| . S ..+=oo|  
| + ..++o |  
| o . +o.. |  
| . o. . |  
| .o |  
+----[SHA256]-----+  
ubuntu@ubuntu:~$ ssh-keygen -t rsa -b 4096  
Generating public/private rsa key pair.
```

```

ubuntu@ubuntu: ~
Enter file in which to save the key (/home/ubuntu/.ssh/id_rsa): Enter passphrase
(empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_rsa
Your public key has been saved in /home/ubuntu/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:41g+my3q8ai+junR8J1maWMwprJRt+4Yjkkq++Qv+gU ubuntu@ubuntu
The key's randomart image is:
+---[RSA 4096]---+
|
|
|
|o E =  S
|oo O = B .
|+. = % +
|.0.0 = =.=
|+=X*O++.+..
+-----[SHA256]-----+
ubuntu@ubuntu:~$
ubuntu@ubuntu:~$ ls -la .ssh
total 16
drwx----- 2 ubuntu ubuntu 120 Jul 30 00:02 .

```

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README.md Update README.md

README

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This is for handsonactivity 2

```

keliste@Keliste-Workstation:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/keliste/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/keliste/.ssh/id_rsa
Your public key has been saved in /home/keliste/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:d5NCOo+ZMSf3l1LmItDyFLIVqvm3qBuGbdSPIWNRlPE keliste@Keliste-Workstation
The key's randomart image is:
+----[RSA 4096]-----+
|      .+o      |
|      ...      |
|      .. E      |
|      oo .      |
|      S=+o+.     |
|      =Xo==..o   |
|      .*=0ooo+ . |
|      o=0B.B o.  |
|      +=0* =.   |
+----[SHA256]-----+
keliste@Keliste-Workstation:~$
keliste@Keliste-Workstation:~$ ls -la ~/.ssh
total 24
drwx----- 2 keliste keliste 4096 Jul 23 10:17 .
drwxr-x--- 15 keliste keliste 4096 Jul 23 09:33 ..
-rw----- 1 keliste keliste 3389 Jul 23 10:17 id_rsa
-rw-r--r-- 1 keliste keliste 753 Jul 23 10:17 id_rsa.pub
-rw----- 1 keliste keliste 1956 Jul 23 09:34 known_hosts
-rw----- 1 keliste keliste 1120 Jul 23 09:34 known_hosts.old
keliste@Keliste-Workstation:~$

```



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```

iquin@Keliste-Workstation: /home/keliste
--expiry-date      value is an expiry date

other
-z, --null          terminate values with NUL byte
--name-only        show variable names only
--includes         respect include directives on lookup
--show-origin      show origin of config (file, standard input, blob, command line)
--show-scope       show scope of config (worktree, local, global, system, command)
--default <value>  with --get, use default value when missing entry

iquin@Keliste-Workstation:/home/keliste$ git config --global user.email qmdiquin@tip.edu.ph
iquin@Keliste-Workstation:/home/keliste$ cat ~/.gitconfig
[user]
    email = qmdiquin@tip.edu.ph
iquin@Keliste-Workstation:/home/keliste$ git config --global user.name MelquisedecIquin
iquin@Keliste-Workstation:/home/keliste$ cat ~/.gitconfig
[user]
    email = qmdiquin@tip.edu.ph
    name = MelquisedecIquin
iquin@Keliste-Workstation:/home/keliste$

```

```
iquin@Keliste-Workstation: /home/keliste
iquin@Keliste-Workstation:/home/keliste$ cat ~/.gitconfig
[user]
  email = qmdiquin@tip.edu.ph
iquin@Keliste-Workstation:/home/keliste$ git config --global user.name MelquisedecIquin
iquin@Keliste-Workstation:/home/keliste$ cat ~/.gitconfig
[user]
  email = qmdiquin@tip.edu.ph
  name = MelquisedecIquin
iquin@Keliste-Workstation:/home/keliste$ which git/usr/bin/git
iquin@Keliste-Workstation:/home/keliste$ git --version
git version 2.34.1
iquin@Keliste-Workstation:/home/keliste$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCAQC+bXHSrfkTqdH3S9vbfxdnk7dXdAigGaPd3jBvcAa4
4qaqNbzwHFWAL/QgO1q0oHWMdStuicYnDqK/NIAT0rt58zdRgFgJEFU0VrbTIbc6cUKZOKzldcdq/jka
LFZ9BWglT2CPw+RbLHM6GAazuVOKZDa7mcBWGqiRjZXxbio6xQNuHneRcf62/NV2rcjmXY/6q0lqQ5Uy
tdRwTBA6cArTGHv37nohHdH96E4MyFwWBb6swcc6vBTy0P1EKsLMV39H8g6cBUm6CUJ+9I8P52QReQ6y
6d8qxvbfZMzhsjjjepZ3jqo1MdtiFGQ1Dq9uPiXTvHn0gXn1x+h6XaX5vmYsCA2L4Vichk8ZB0tpdc8C
Y0bU2buaFLwKkC9WaPFG37RUEhSIU3wmoerYeF4jrny0u4tKwFOSJ5r0zzdJPxlbV6WKooa+WBV2+tbQ
KDqSDVSWFrg4F43nWsbKDNTmNXuNuwpmoWnxmV83/132pyGSzHF2BLZEnsH0iWEFZa0SRdqNKJ/6g9qr
LK9y0t60dxfpoMKxfjnFGLWbSB6lh7k3r/eSy0TAVwKrmLA2R/Svw4R8g5+PpyUGmAkeVtNvTp4JM5b
ZG0s3kPx2WvrDvdWwCYZEWe5FiyQZbgZFhUzmJR17p9Zq5AtQvG12D/B12xtBt60RD7sjsR2RULqWyZB
5Q== iquin@Keliste-Workstation
iquin@Keliste-Workstation:/home/keliste$
```



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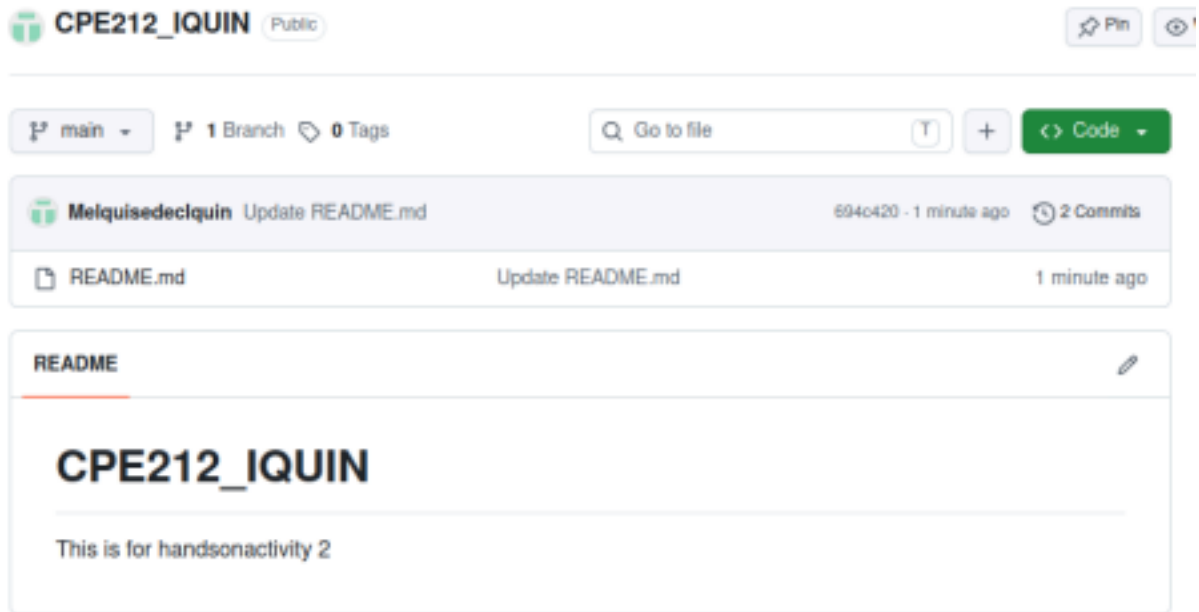
main 1 Branch 0 Tags

MelquisedecIquin Initial commit 75147eb · 1 hour ago 1 Commit

README.md Initial commit 1 hour ago

README

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

- How will you describe the ssh-program? What does it do?
- SSH is a network protocol that allows the user to connect and control a remote computer over an unsecured network. Unlike other protocols, ssh encrypts all communication such as passwords and data.
2. How do you know that you already installed the public key to the remote servers?
- You can check the remote server by running `cat ~/.ssh/authorized_keys` which will display the public key.
3. What sort of things have we so far done to the remote servers using ansible commands?
- Ansible ad-hoc commands to test connectivity such as `ansible all -m ping` that can be used to ping another server or system.
4. How important is the inventory file?
- Inventory file is important because it tells ansible exactly which remote servers to manage.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code

assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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